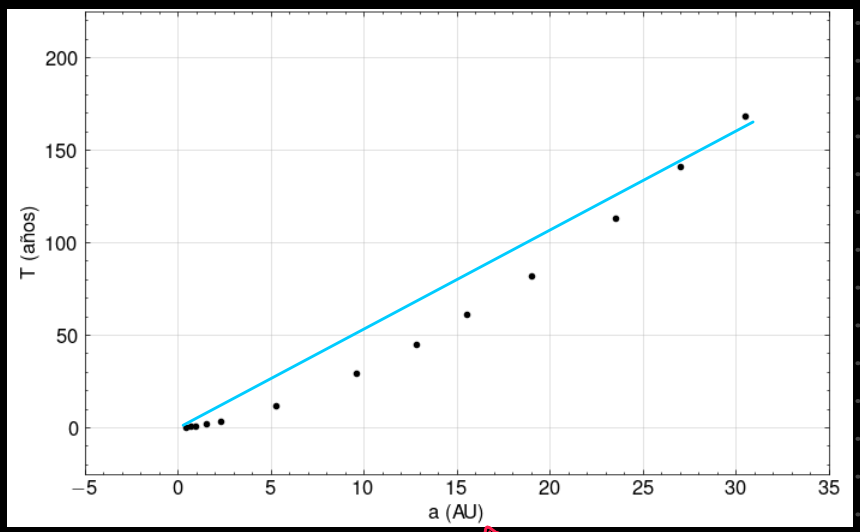
Ejemplo Objetos en un sistema planetario

a (AU)	T (yr)
0.41	0.27
0.70	0.60
0.98	0.99
1.55	1.95
2.30	3.55
5.25	11.70
9.60	29.50
12.80	45.00
15.50	61.00
19.0	82.00
23.5	113.00
27.0	141.00
30.5	168.00



variable unidades

→ Tercera Ley de Kepler $T^2 \propto a^3$

$T^2 = k\, a^3$

$y = k\, x$ ¡ecuación de una recta!

→ $x_i \rightarrow a^3$

→ $y_i \rightarrow T^2$

	x_i (a^3)	y_i (T^2)	x_i^2	$x_i y_i$
1	0.07	0.07	0.005	0.01
2	0.34	0.36	0.118	0.12
3	0.94	0.98	0.886	0.92
4	3.72	3.80	13.807	14.16
5	12.71	12.60	148.036	153.33
6	144.70	136.89	20938.994	19808.41
7	884.74	870.25	782757.790	769941.50
8	2097.15	2025.00	4398046.51	4246732.80
9	3723.88	3721.00	13867245.00	13856538.90
10	6859.00	6724.00	47045881.00	46119916.00
11	12977.88	12769.00	168425240.00	165714486.00
12	19683.00	19881.00	387420489.00	391317723.00
13	28372.62	28224.00	805005849.00	800700968.00

$\sum x_i = 74760.21$ $\sum y_i = 74368.99$ $\sum x_i^2 = 1426966610.13$ $\sum x_i y_i = 1422834283.01$

$\bar{x} = 5750.79$ $\bar{y} = 5720.69$

- $m = \dfrac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2}$
- $b = \bar{y} - m\bar{x}$

$m = 0.99 \qquad b = 27.41$

$$\boxed{y = 0.99\,x + 27.41}$$

$T^{-2} \qquad a^3$

$$T^2 = k\,a^2$$

$$\boxed{k = 0.99}$$

$$T^2 = \underbrace{\frac{4\pi^2}{GM}}_{k}\,a^3$$

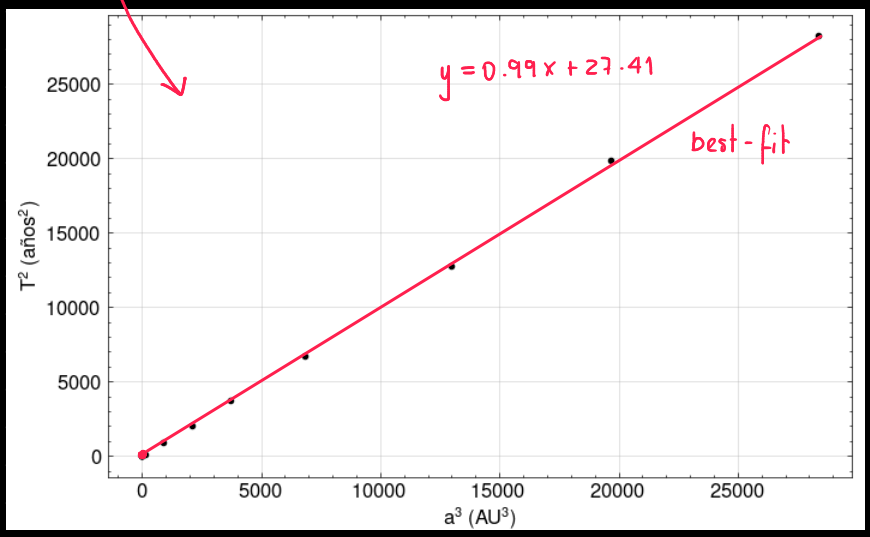
cuando utilizamos

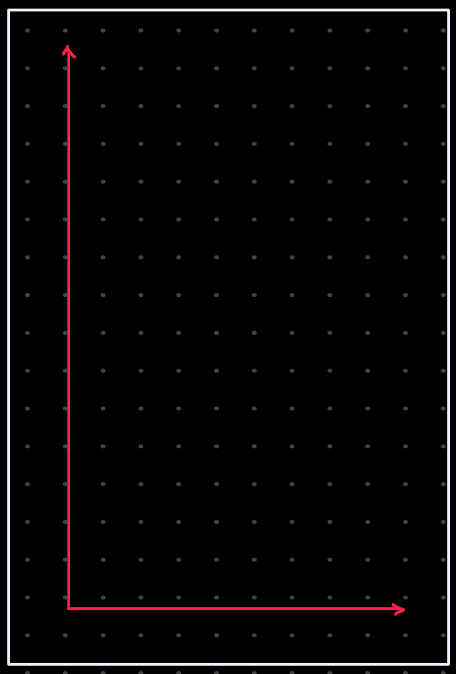
- años
- unidades astronómicas
- masas solares

$$4\pi^2 = GM$$

$$\frac{4\pi^2}{GM} = 1$$

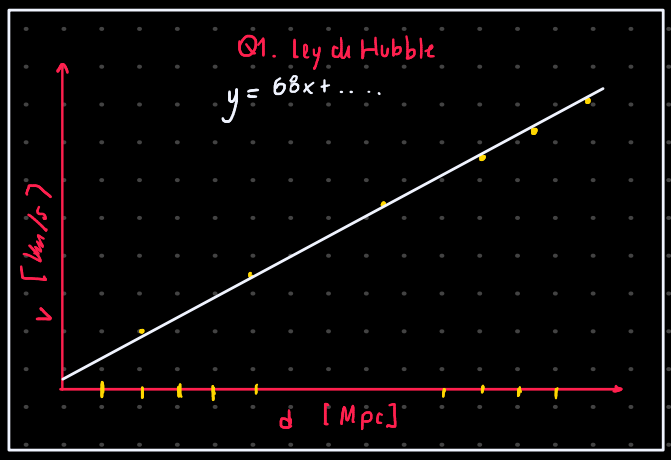
$$T^2 = a^3$$





UTILIZAR TODO EL ESPACIO

regresión lineal → línea best-fit



0 100

Análisis de errores (incertidumbres)

Ningún instrumento de medición es perfecto, todas las mediciones llevan cierto error

1. Error sistemático (consistentes, reparables)
2. Error aleatorio

	error
Instrumento	+ 0.02
corrección	− 0.02

error	corrección
0.11	0.09
0.12	0.10
0.14	0.12

regla. 1 mm

medición: 4.123 m (mm)

no podemos medir

instrumento → incertidumbre asociada

mediciones

$$X = X_{mes} \pm \Delta x$$

incertidumbre absoluta

mejor estimación

$$X_{mes} - \Delta x < X_{mes} < X_{mes} + \Delta x$$

Incertidumbre absoluta vs. relativa

- Incertidumbre absoluta

$$d = \underset{x}{50} \pm \underset{\Delta x}{2} \text{ Mpc}$$

[Mpc]

- Incertidumbre relativa

$$\frac{\Delta x}{x}, \quad \frac{2}{50} = 0.04$$

$$\frac{\Delta x}{x} \cdot 100\% , \quad \frac{2}{50} \cdot 100\% = 4\%$$

Origen de las incertidumbres

1. Resolución del instrumento — regla (1 mm) — $\Delta x = \pm 0.5$ mm

2. mediciones repetidas

distancia (pc)

1 estrella

10
11
13
10
9

$\sum d_i = 53$

$\bar{d} = 10.6$

$d = \bar{d} \pm \sigma$

$\boxed{d = 10.6 \pm 1.36}$

$\sigma = \sqrt{\frac{\sum d_i^2}{n} - \bar{d}^2}$

$\sigma = 1.36$

3. juicio / lectura

Propagación de errores

$x + \Delta x, \quad y + \Delta y$

1. suma o resta

→ ideal $z = x + y$

$\Delta z = \Delta x + \Delta y$

(siempre se suman)

reportar

$z + \Delta z$

$z = (x+y) \pm (\Delta x + \Delta y)$

$z = (x-y) \pm (\Delta x + \Delta y)$

2. Multiplicación

$z = x \cdot y$

$x = x_0 \pm \Delta x$

$y = y_0 \pm \Delta y$

$\Delta z = z\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right)$

$z = (x \cdot y) \pm z\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right)$

$x = 2 \pm 0.1 \qquad y = 4 \pm 0.2$

$z = (x \cdot y) \pm (x \cdot y)\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right)$

$z = (2 \cdot 4) \pm (2 \cdot 4)\left(\frac{0.1}{2} + \frac{0.2}{4}\right)$

$z = 8 \pm (2.4 \cdot 0.1) \longrightarrow \boxed{z = 8 \pm 0.24}$

3. división

$$z = \frac{x}{y} \qquad \Delta z = z\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right)$$

4. potencias

$$z = x^{n} \qquad \Delta z = z \cdot n \left(\frac{\Delta x}{x}\right)$$

Reportar resultados

Regla 1: igualar la precisión

$$V = 6723 \pm 241$$

$$V = (6.723 \times 10^{3} \pm 0.241 \times 10^{3})$$

$$V = (6.7 \pm 0.2)\, 10^{3}$$

Regla 2: el valor y la incertidumbre deben alinearse

mal 6.728 ± 0.2 bien 6.7 ± 0.2